HYDRIS AI

HYDRIS FIELD GLASSES · PRODUCT BOOK, PART 8

Hey Hydris voice assistant

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HFG-BOOK-08	1.0	Draft for review	October 2026

Document control

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1 Hey Hydris voice assistant

1.1 Purpose and problem

An operator at a clarifier with gloves on cannot scroll a dashboard. "Hey Hydris" lets them ask out loud ("what's the outlet pH?", "any open alarms?", "what are my tasks?") and hear a short answer on the glasses speakers or the phone, with the facts behind it on screen. Every answer comes from the caller's own plant data, never from general knowledge, and the assistant cannot change anything.

5
Read-only tools
readings, alarms, my work, assets, SOPs

2
Answer engines
Claude, or the rules engine

0
Write tools
advisory only

0
Invented numbers allowed
target and design rule

1.2 Users and jobs

Table 1. Users and jobs

User	Job
Field operator	Get a reading, its age and alarm state without touching the phone
Maintenance engineer	Hear my open work and the procedure for an asset
Supervisor	Check open alarms on my sites while walking the plant

1.3 Scope

Table 2. In scope and out of scope

In scope	Out of scope
Wake phrase, spoken question, spoken answer	Operating equipment or changing records
Readings, alarms, my work, asset status, approved SOPs	Answers from outside the tenant's data
Follow-up questions with history	Indian languages (Planned, FR-073)
Glasses, phone and browser speech	"Hey Meta, start Hydris" as a supported path (experimental)

1.4 Wake word flow

The wake phrase is detected on the device with a pattern tolerant of how recognizers hear "Hydris" ("hydras", "hi dris", "idris"). After an answer the assistant keeps listening for a follow-up; "stop", "cancel" or "thanks" ends it.

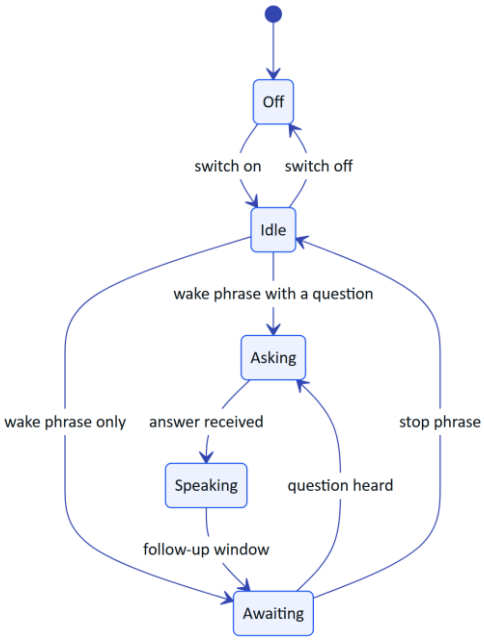


Figure 1. *Wake phrase states on the phone*

> **NOTE**
> The glasses web viewport closes the follow-up window after 8 s of silence; the Android app keeps it open until a stop phrase. Aligning them on 8 s is Planned.

1.5 Speech engines

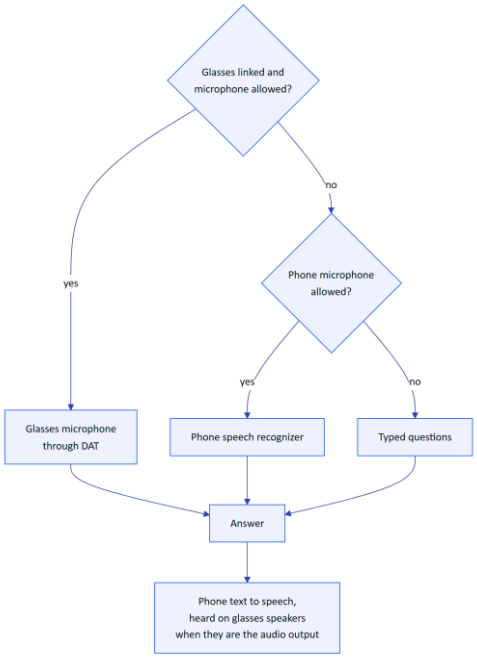


Figure 2. *Which microphone and speaker are used*

Table 3. Speech engines by surface

Surface	Listens with	Speaks with	Status
Android app, glasses	DAT Speech on the glasses microphone	Phone text to speech over Bluetooth	Partial: not yet run on hardware
Android app, phone	Android speech recognizer, restarted after each result	Phone text to speech	Built
Glasses web viewport and simulator	Bridge first, then the browser Web Speech API	Browser speech synthesis	Built

Transcripts are dropped while the assistant is speaking, so it does not answer itself.

1.6 Grounded answering

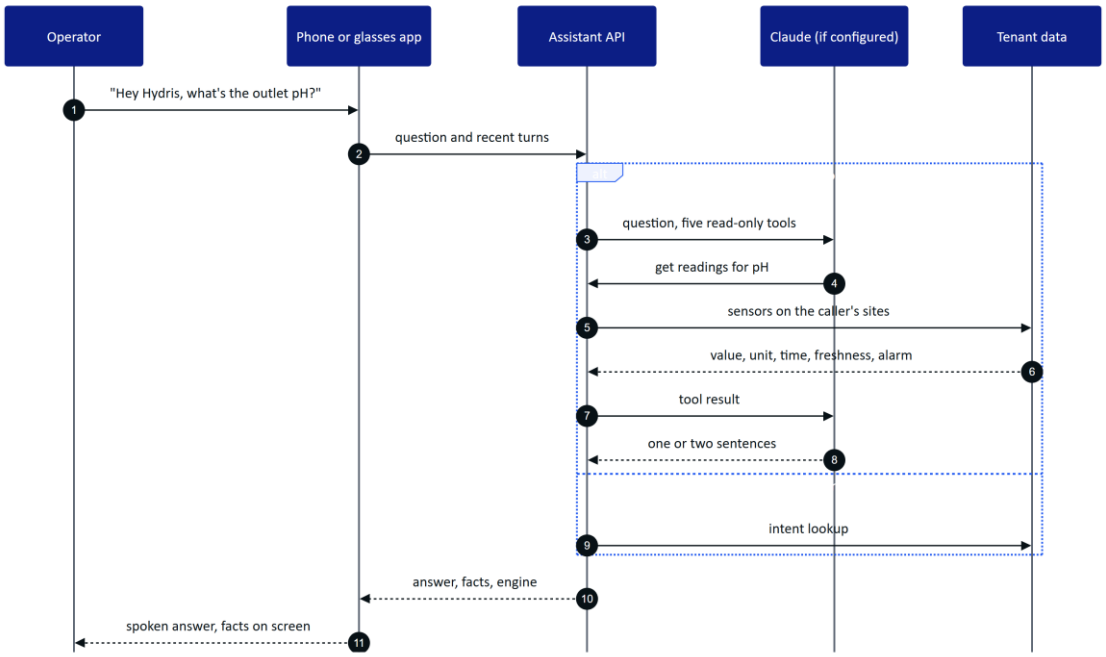


Figure 3. *A spoken question answered from the caller's own data*

When an Anthropic credential is configured, Claude answers with up to five rounds of read-only tool calls. Otherwise, and whenever the model fails, a deterministic rules engine answers readings, alarms, my work, asset status, procedures and help, and says "I can't answer that from your plant data" for anything else. Every tool runs with the caller's row-level and site scope, so the assistant can only say what the caller could see in the console.

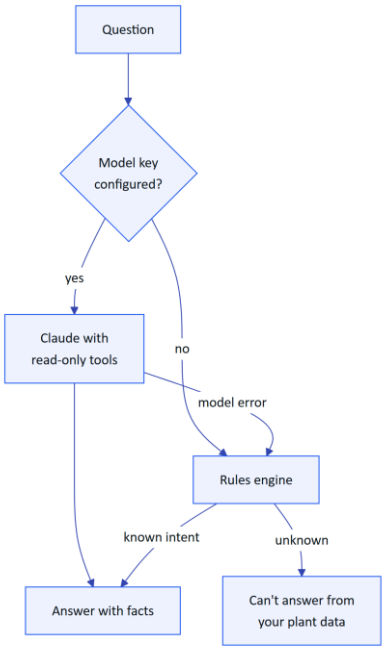


Figure 4. *Choosing the engine for each question*



Figure 5. *The assistant answering with the facts behind it*

1.7 Guardrails

- **Advisory only.** The tool list holds only read tools and the system prompt forbids operating equipment (FR-078).
- **Never invents numbers.** Answers return facts (label, value, unit, time, source); the screen shows up to four under the answer. Rules answers contain only numbers that appear in facts.
- **States the data age.** Answers say "measured N minutes ago"; stale readings add "not current, check the sensor"; missing data is "no data", never zero (NFR-036).
- **Labels synthetic data.** Demo answers mark their facts as synthetic demo data (FR-075).
- **Approved documents only.** Draft SOPs never reach an answer (FR-111).

Table 4. Example dialogue (illustrative of the rules described above, not a recorded transcript)

Operator says	Hydris answers	Facts shown
"Hey Hydris, what's the outlet pH?"	Outlet pH is 7.3, measured a few minutes ago, within range	7.3 · Outlet pH
"And the inlet?"	Answers for the inlet, using the conversation history	Inlet pH value and time
"Any open alarms?"	Counts open alarms on my sites and names the most severe	Alarm titles and states
"What are my tasks?"	Lists my open maintenance tasks and due inspections	Task titles
"What's the weather tomorrow?"	I can't answer that from your plant data	None
"Thanks"	Ends the follow-up window	None

1.8 Requirements

Table 5. Voice assistant requirements

ID	Requirement	Pri	Status
FR-070	Grounded SOP retrieval from approved documents with a safety notice	M	Built
FR-071	Wake phrase and short spoken answers	M	Built
FR-072	Answers only from tenant data through scoped tools; rules fallback never guesses	M	Built
FR-073	Hindi, Tamil, Marathi, Gujarati	S	Planned (Advanced)
FR-074	Glasses microphone with phone fallback; answers on glasses speakers	M	Partial: glasses not run on hardware
FR-075	Facts with source; synthetic data labelled	M	Built
FR-076	Follow-ups with history (up to 8 turns)	S	Built
FR-077	"Hey Meta, start Hydris" opens the assistant	C	Partial (experimental)
FR-078	No write tools	M	Built

1.9 Success metrics

Table 6. Voice targets (targets, not measured results)

Metric	Pilot target
Questions answered with facts, not "can't answer"	85 % or more
Numbers in answers missing from facts	0
Spoken answer after the question ends	Under 3 s on 4G

1.10 Status, gaps and next steps

- **Logging for metrics.** Engine and intent logging on the assistant is Planned for the pilot; until then the targets cannot be measured.
- **Glasses microphone on hardware** is the first hardware run item for Q4 2026.
- **Wake phrase false accepts** are tested against eight hours of plant conversation before tightening the pattern.

- **Model processing consent.** A tenant must agree to Claude processing before a pilot uses it; otherwise the rules engine runs alone.
- **Languages.** Indian-language questions and answers depend on server support and are Planned.